

AMR rules in action

Sixty moments from one five-robot run, each photographed the instant a condition fired. The caption on every picture was written at that moment: the condition, what it means, and where all five robots were.

What was seen

Condition	Times seen
machine has not granted entry	12
dwelling at the port	11
exit complete	6
rule 1 lower number gives way	5
stand aside begun	5
arrived	4
passer waiting for the yielder to stand aside	4
stand aside complete	3
junction held by another robot	3
stood aside waiting to be passed	3
encounter over	3
layer 1 on course to touch another robot	1

What was NOT seen

These conditions exist in the code and did not fire once during the run. They are the failure paths that dominated earlier runs, so their absence is the result, not a gap in the recording.

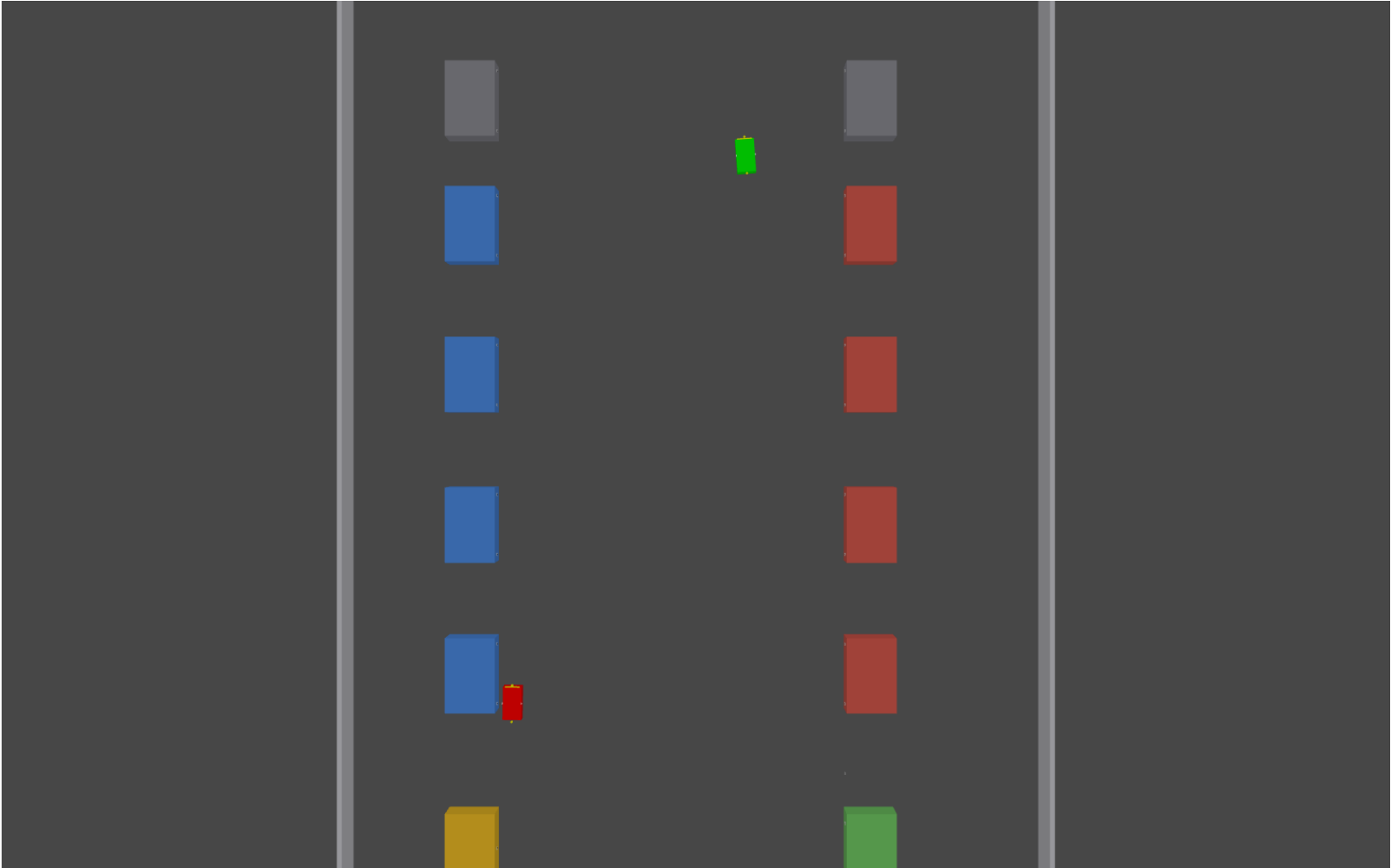
- ✓ **gave way for** — the 45-second give-up that cost jobs in earlier runs
- ✓ **deadlocked with** — the deadlock breaker
- ✓ **standing aside threat** — a yielder blocked on its way to the lay-by
- ✓ **nowhere to stand aside** — a yielder with no reachable lay-by
- ✓ **robot ahead on the road** — the protective corridor stopping a robot
- ✓ **queueing behind** — a follower held behind a robot that is giving way
- ✓ **pulling out road not clear** — a robot unable to reverse out of a bay

✓ **battery flat** — a robot stopped with a flat battery

machine has not granted entry (1 of 12)



machine has not granted entry (2 of 12)



amr1: machine has not granted entry

The machine has not opened its door. Nothing to do with traffic.

amr1	(-13.44, 7.67)	GRV1_LD	machine has not granted entry
amr2	(13.09, -3.60)	CTR1_LD	moving
amr3	(29.08, -1.50)	-	moving
amr4	(29.08, -3.65)	-	moving
amr5	(26.11, -3.82)	SLT_LD1	moving

17:15:59 STATE amr1(-13.4,+7.7)->GRV1_LD d=1.1 v=0.02 [machine has not granted entry] | amr2(+13.2,-3.6)-

machine has not granted entry (3 of 12)



machine has not granted entry (4 of 12)



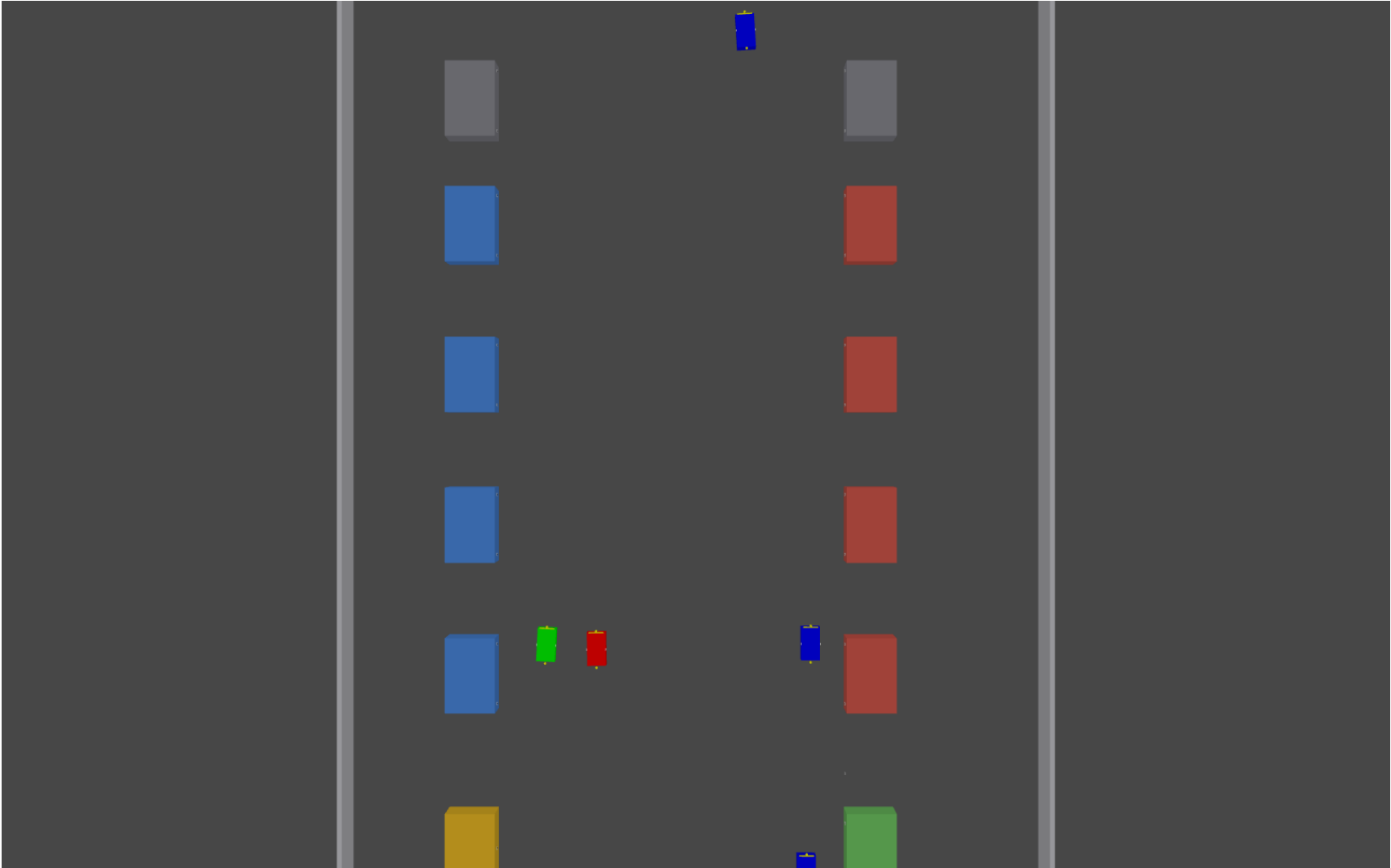
machine has not granted entry (5 of 12)



machine has not granted entry (6 of 12)



machine has not granted entry (7 of 12)



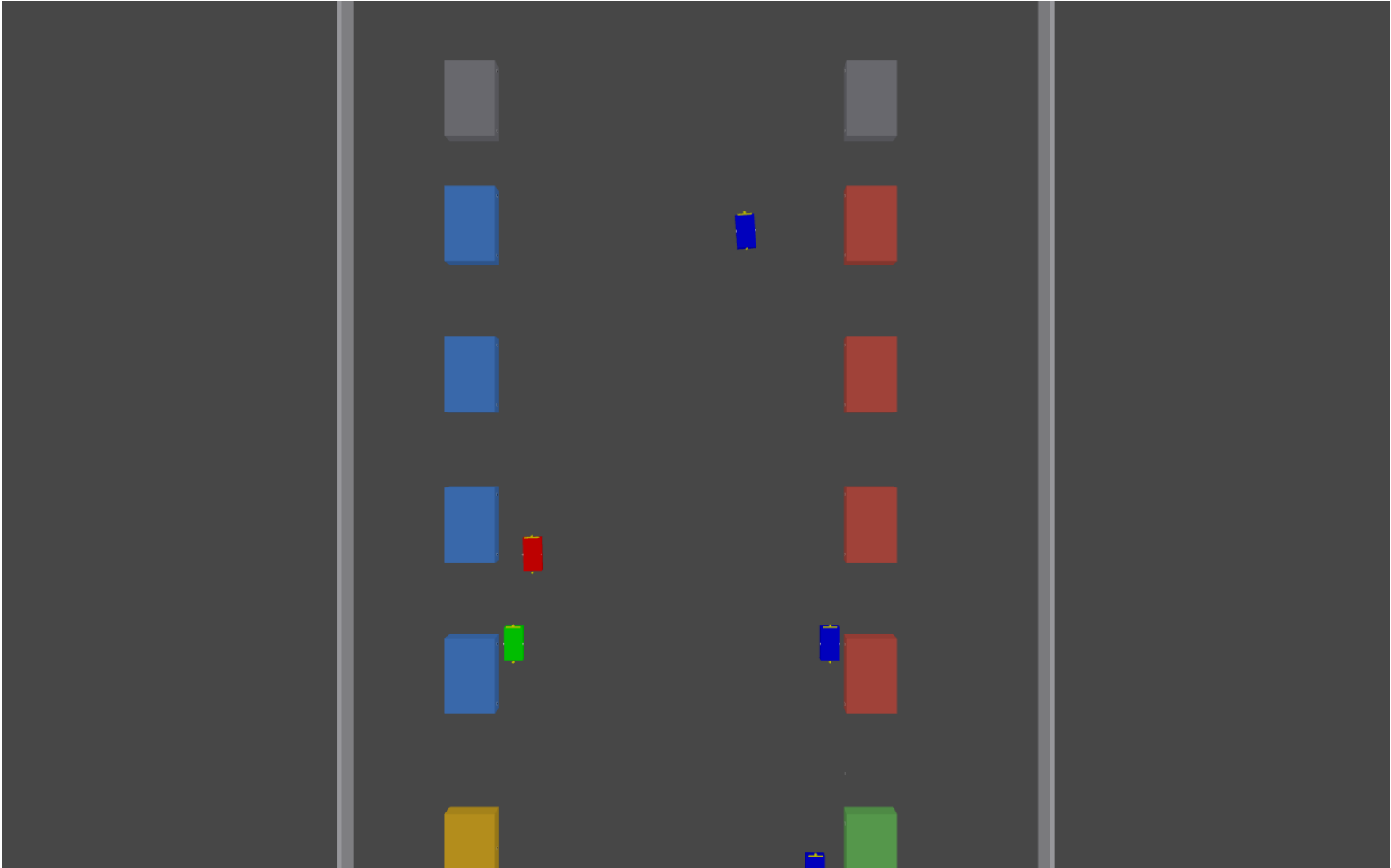
amr4: machine has not granted entry

The machine has not opened its door. Nothing to do with traffic.

amr1	(-10.92, 3.60)	GRV2_LD	layer 1: on course to touch another robot
amr2	(-10.65, 5.78)	GRV1_ULD	layer 1: on course to touch another robot
amr3	(19.23, -3.60)	SLT_LD3	moving
amr4	(-21.59, -6.53)	SLT_LD2	machine has not granted entry
amr5	(-10.57, -6.71)	CTR1_ULD	machine has not granted entry

17:18:55 STATE amr1(-10.9,+3.6)->GRV2_LD d=4.7 v=0.57 | amr2(-10.6,+5.8)->GRV1_ULD d=0.8 v=0.49 | amr3(+1

machine has not granted entry (8 of 12)



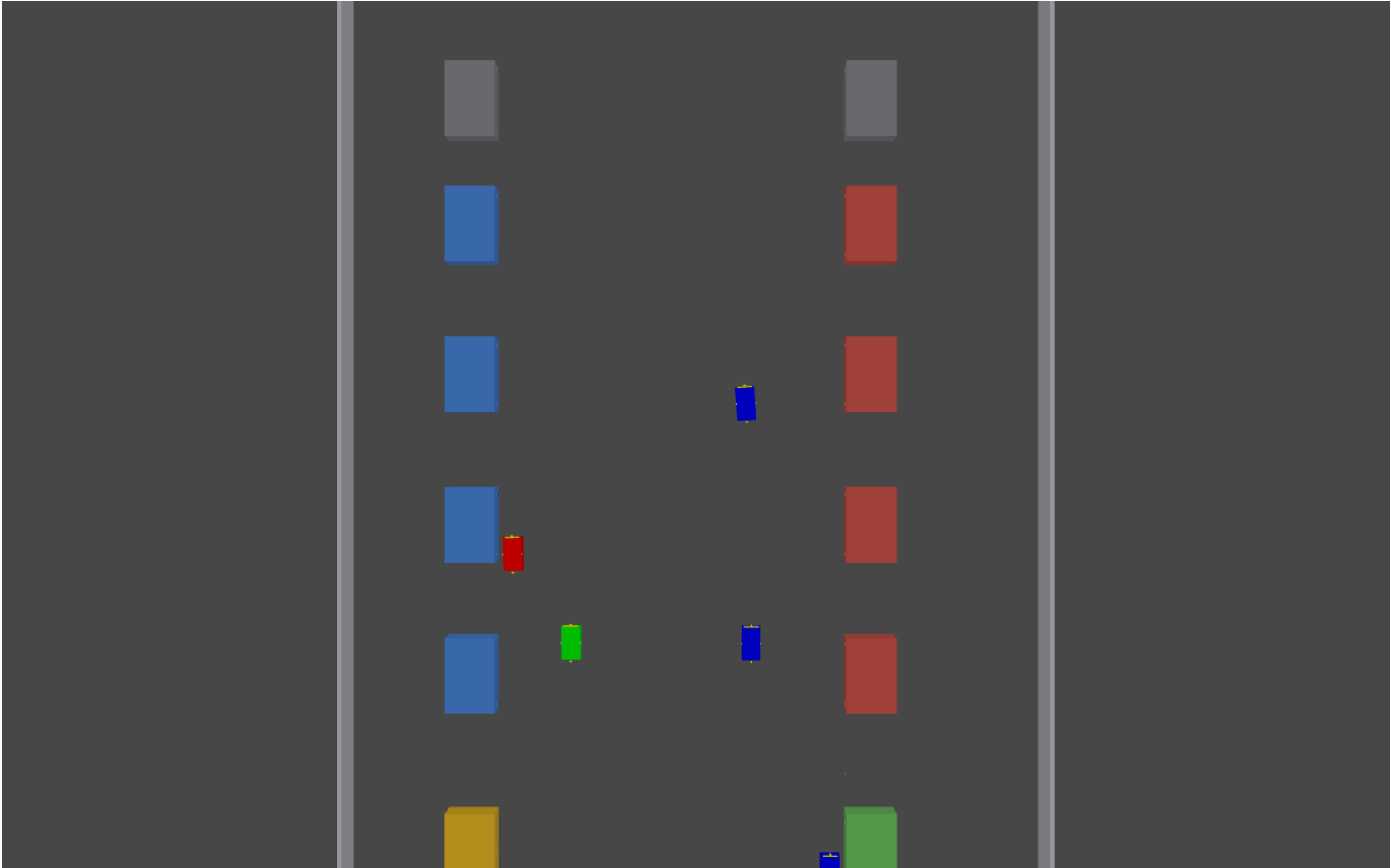
amr5: machine has not granted entry

The machine has not opened its door. Nothing to do with traffic.

amr1	(-6.25, 6.65)	GRV2_LD	machine has not granted entry
amr2	(-10.56, 7.59)	GRV1_ULD	layer 1: on course to touch another robot
amr3	(9.57, -3.60)	SLT_LD3	moving
amr4	(-21.60, -6.93)	SLT_LD2	machine has not granted entry
amr5	(-10.56, -7.66)	CTR1_ULD	machine has not granted entry

17:19:12 STATE amr1(-6.2,+6.6)->GRV2_LD d=0.0 v=0.24 | amr2(-10.6,+7.6)->GRV1_ULD d=1.0 v=0.04 | amr3(+9.

machine has not granted entry (9 of 12)



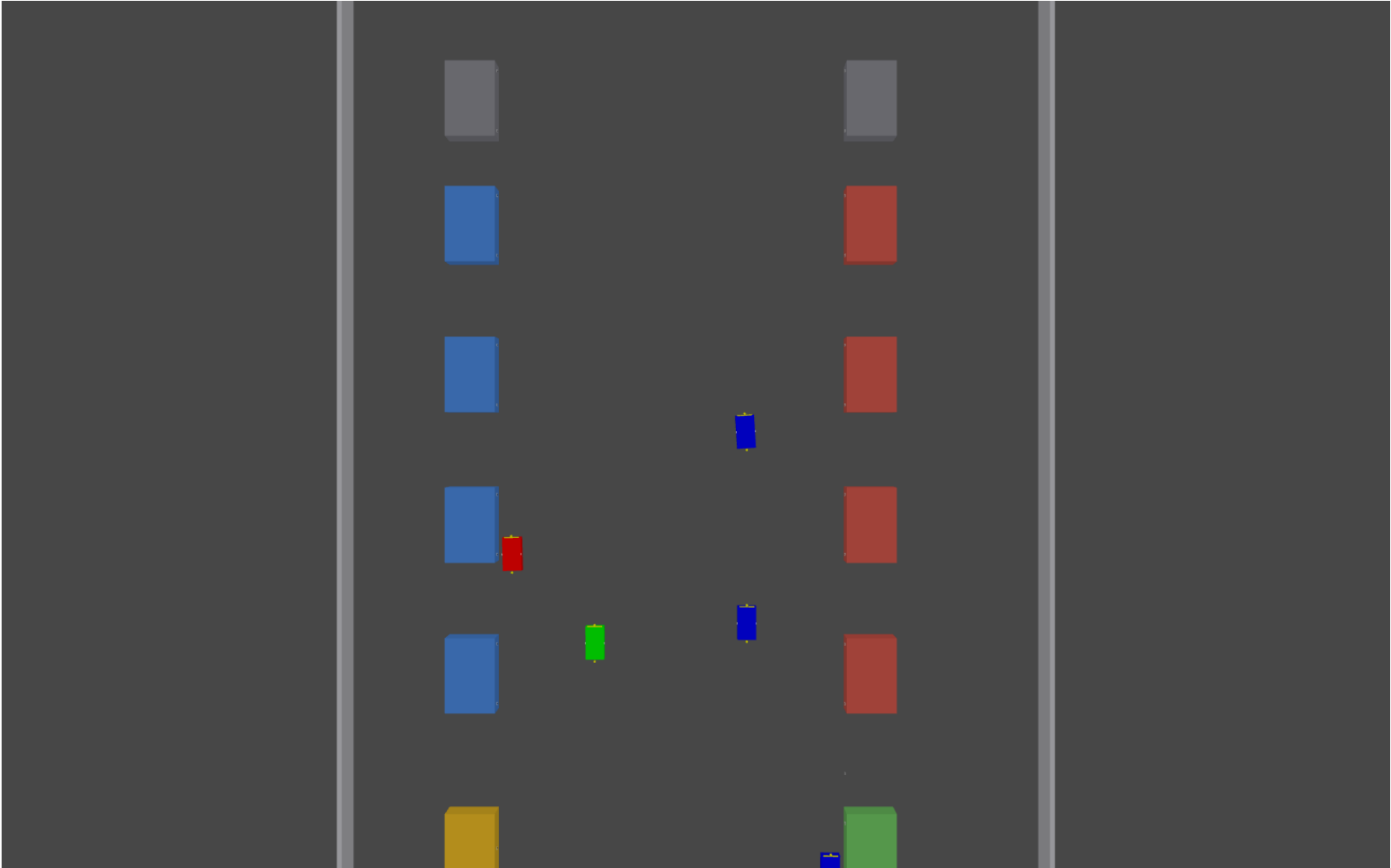
amr4: machine has not granted entry

The machine has not opened its door. Nothing to do with traffic.

amr1	(-6.24, 7.63)	GRV2_LD	machine has not granted entry
amr2	(-10.51, 5.02)	-	arrived
amr3	(1.19, -3.60)	SLT_LD3	moving
amr4	(-21.60, -7.67)	SLT_LD2	machine has not granted entry
amr5	(-10.54, -4.11)	-	arrived

17:19:26 STATE amr1(-6.2,+7.6)->GRV2_LD d=1.0 v=0.03 | amr2(-10.5,+5.2)->None d=0.0 v=0.60 | amr3(+1.3,-3

machine has not granted entry (10 of 12)



amr1: machine has not granted entry

The machine has not opened its door. Nothing to do with traffic.

amr1	(-6.24, 7.67)	GRV2_LD	machine has not granted entry
amr2	(-10.56, 3.74)	-	exit complete
amr3	(-0.16, -3.60)	SLT_LD3	moving
amr4	(-21.60, -7.69)	SLT_LD2	machine has not granted entry
amr5	(-9.86, -3.66)	-	exit complete

17:19:28 STATE amr1(-6.2,+7.7)->GRV2_LD d=1.1 v=0.02 [machine has not granted entry] | amr2(-10.6,+3.9) i

machine has not granted entry (11 of 12)

amr3: machine has not granted entry

The machine has not opened its door. Nothing to do with traffic.

amr1	(-20.40, 7.71)	ASRS	exit complete
amr2	(-15.53, -1.63)	CTR3_LD	junction held by another robot
amr3	(-19.20, -7.68)	SLT_LD3	machine has not granted entry
amr4	(-3.36, -7.40)	CTR2_ULD	machine has not granted entry
amr5	(-16.37, -3.60)	SLT_LD1	passer: waiting for the yielder to stand aside

17:20:47 STATE amr1(-20.4,+7.7)->ASRS d=1.1 v=0.01 [exit complete] | amr2(-15.5,-1.6)->CTR3_LD d=2.9 v=0.

machine has not granted entry (12 of 12)



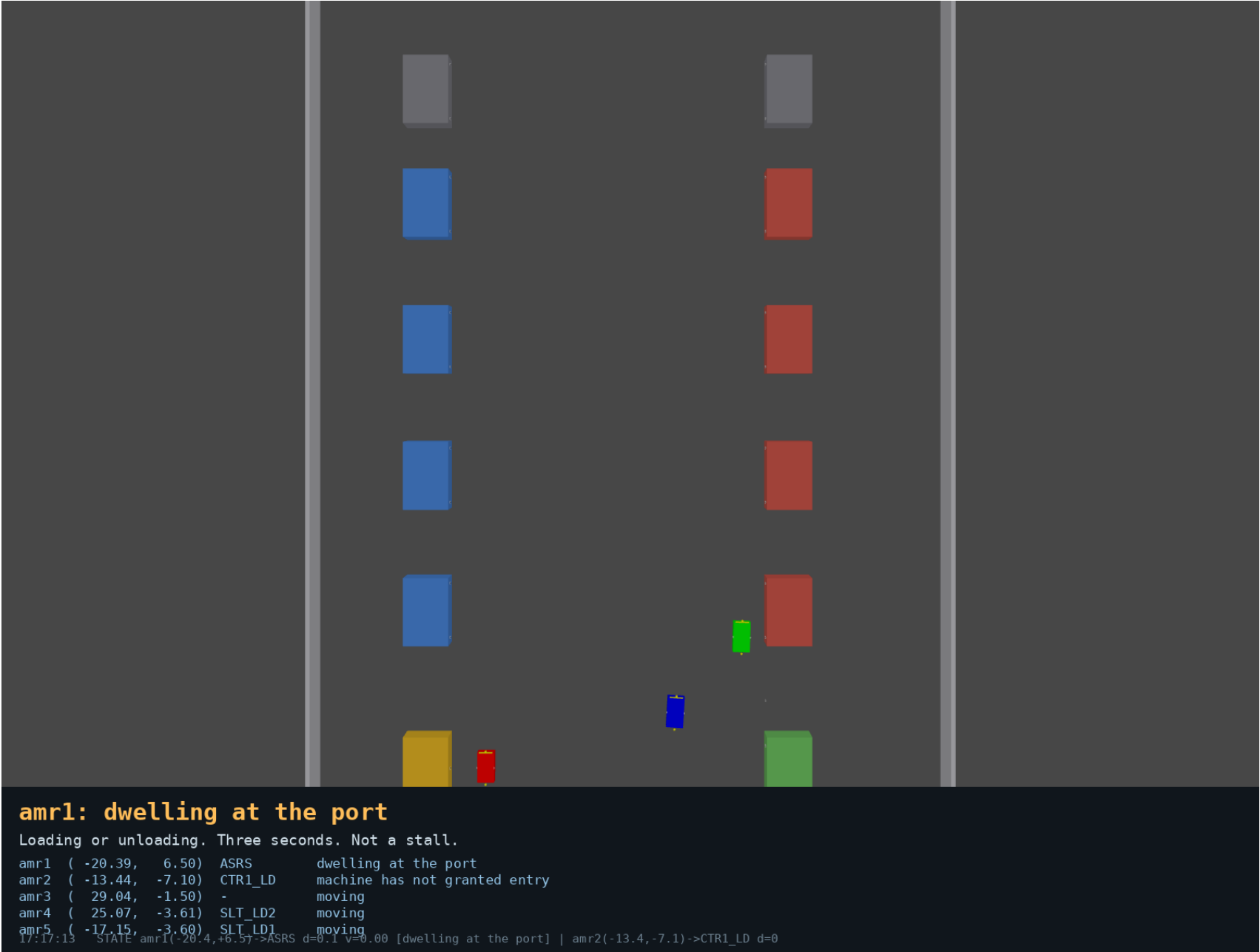
dwelling at the port (1 of 11)



dwelling at the port (2 of 11)



dwelling at the port (3 of 11)



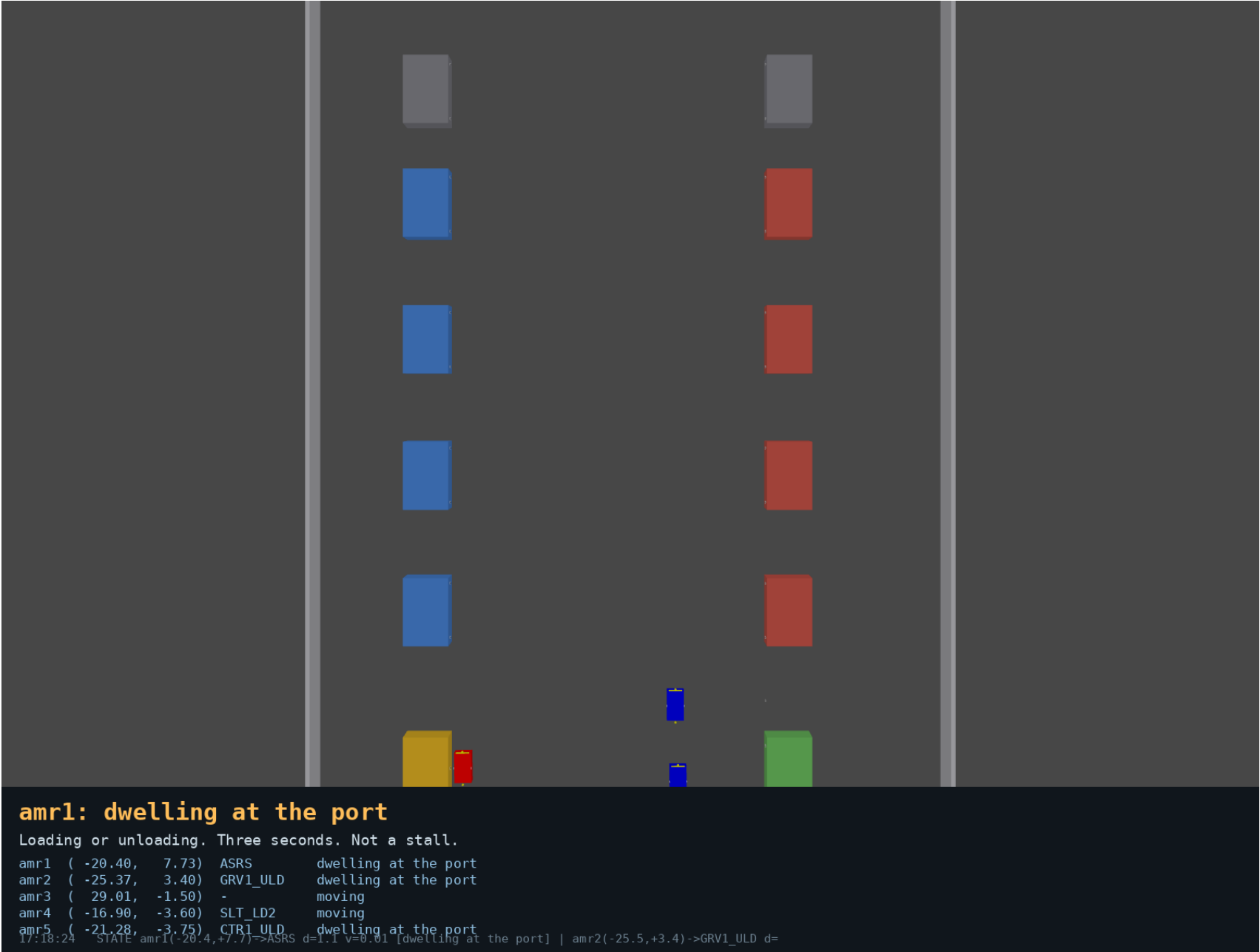
dwelling at the port (4 of 11)



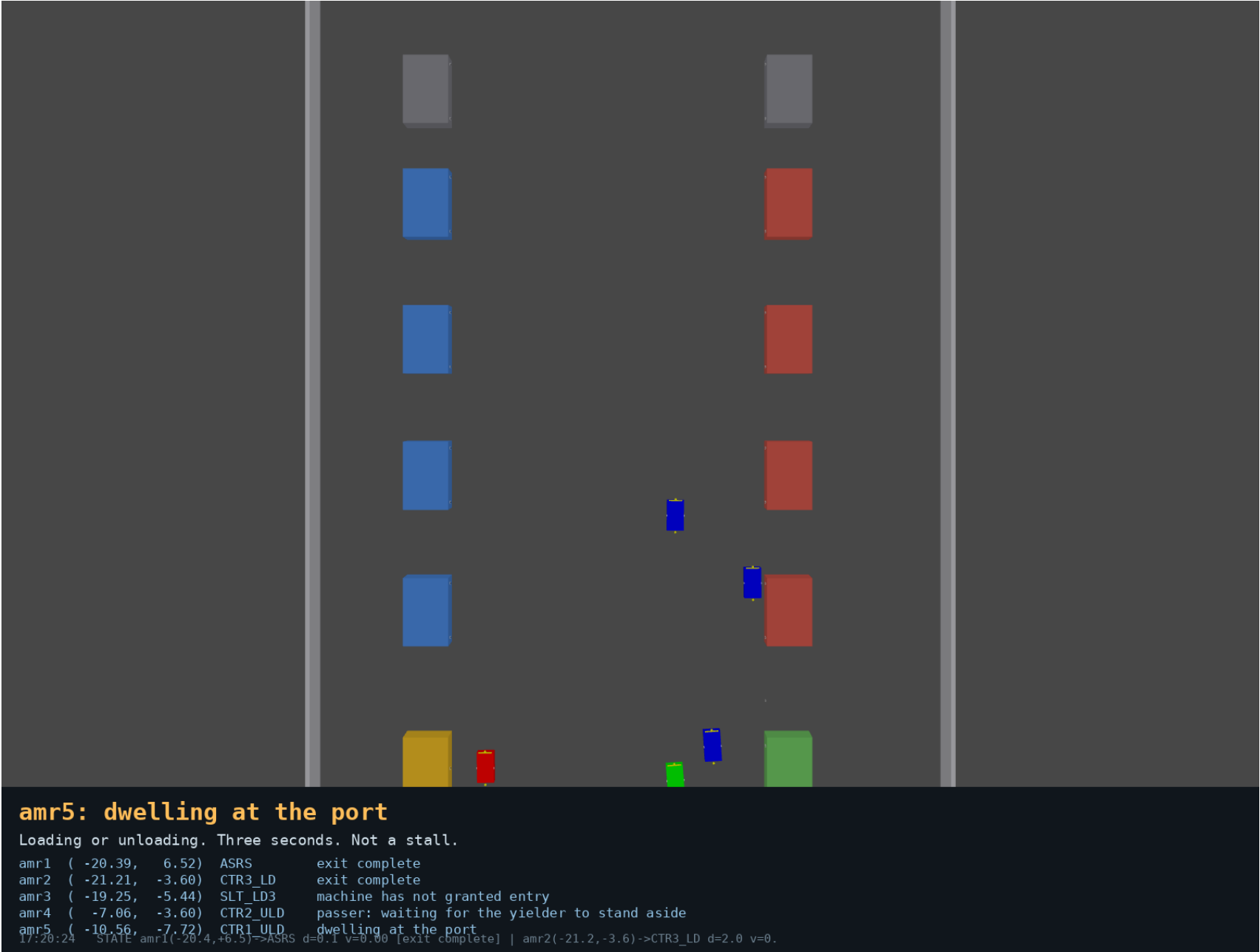
dwelling at the port (5 of 11)



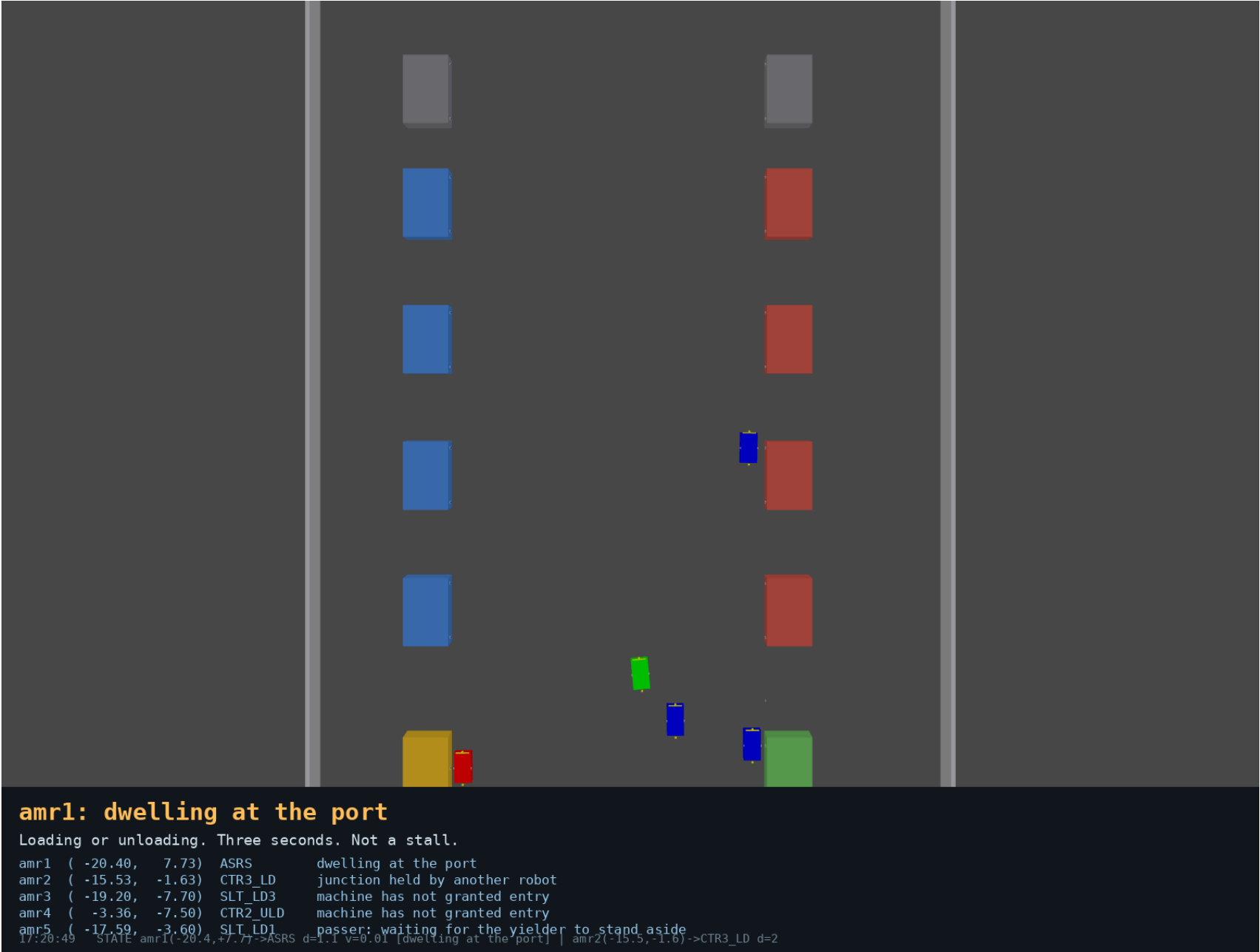








dwelling at the port (10 of 11)





arrived (1 of 4)

arrived (2 of 4)



arrived (3 of 4)

amr5: arrived

At its goal.

amr1	(-6.24, 7.41)	GRV2_LD	machine has not granted entry
amr2	(-10.56, 7.72)	GRV1_ULD	layer 1: on course to touch another robot
amr3	(4.73, -3.60)	SLT_LD3	moving
amr4	(-21.60, -7.53)	SLT_LD2	machine has not granted entry
amr5	(-10.55, -7.69)	-	arrived

17:19:20 STATE amr1(-6.2,+7.4)->GRV2_LD d=0.8 v=0.08 | amr2(-10.6,+7.7)->GRV1_ULD d=1.1 v=0.01 [layer 1:



exit complete (1 of 6)



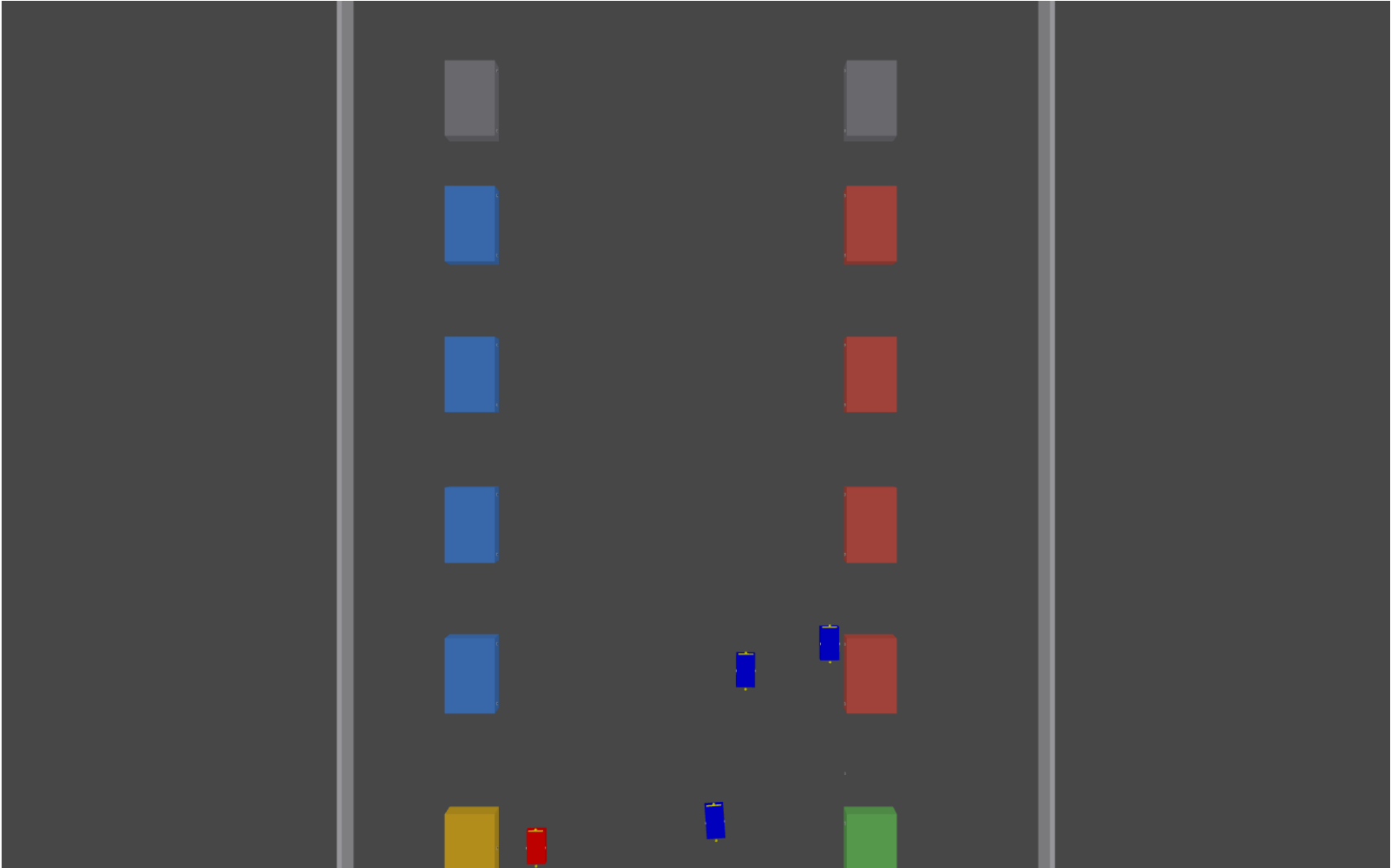




exit complete (4 of 6)



exit complete (5 of 6)



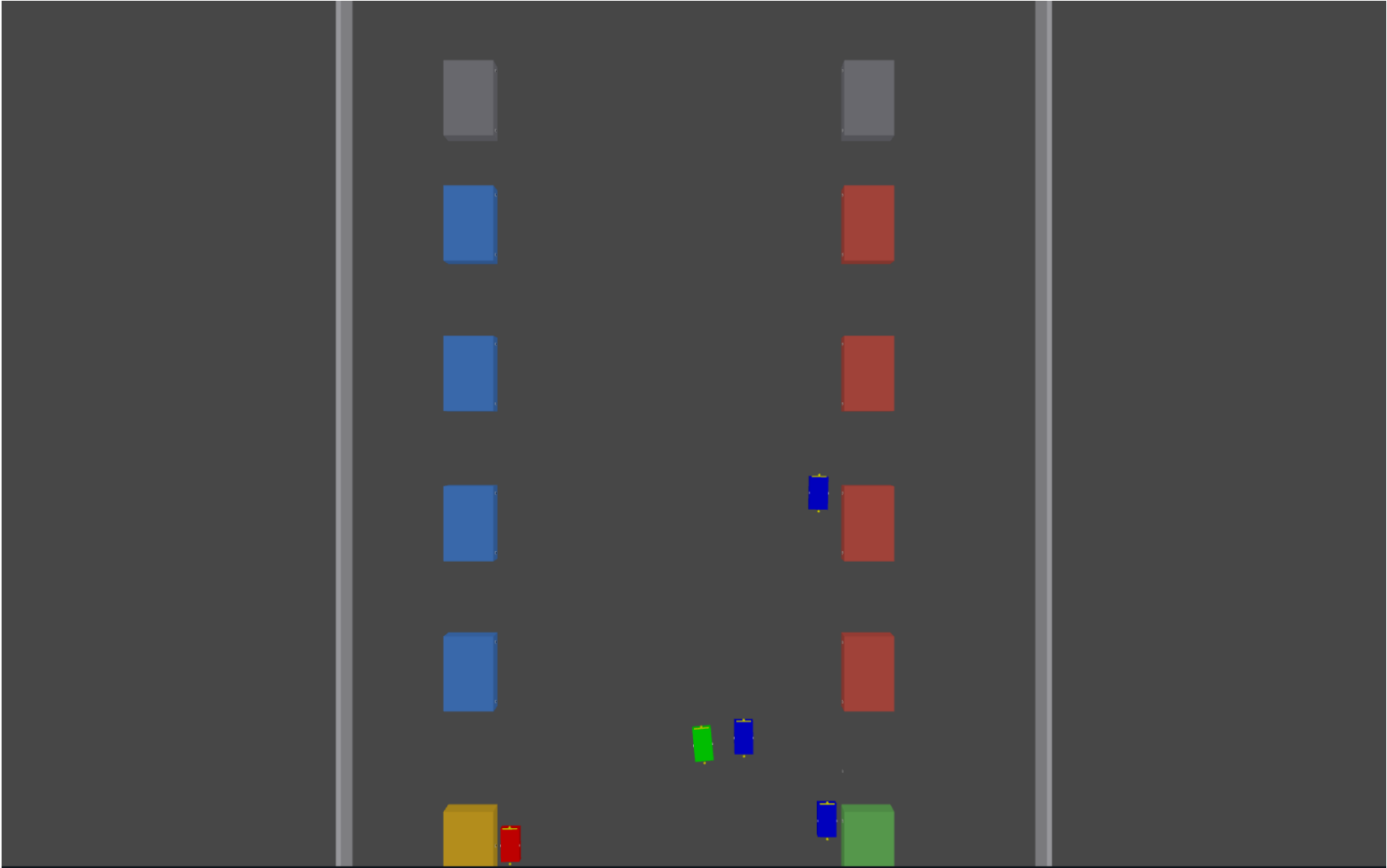
amr5: exit complete

Clear of the bay and the interlock is released. The next robot may come in.

amr1	(-20.39, 6.51)	ASRS	exit complete
amr2	(-26.34, -3.32)	CTR3_LD	exit complete
amr3	(-19.04, -2.12)	SLT_LD3	junction held by another robot
amr4	(-11.86, -3.60)	CTR2_ULD	passer: waiting for the yielder to stand aside
amr5	(-10.56, -7.65)	CTR1_ULD	exit complete

17:20:14 STATE amr1(-20.4,+6.5)=>ASRS d=0.1 v=0.00 [exit complete] | amr2(-26.3,-3.3)->CTR3_LD d=2.4 v=0.

exit complete (6 of 6)

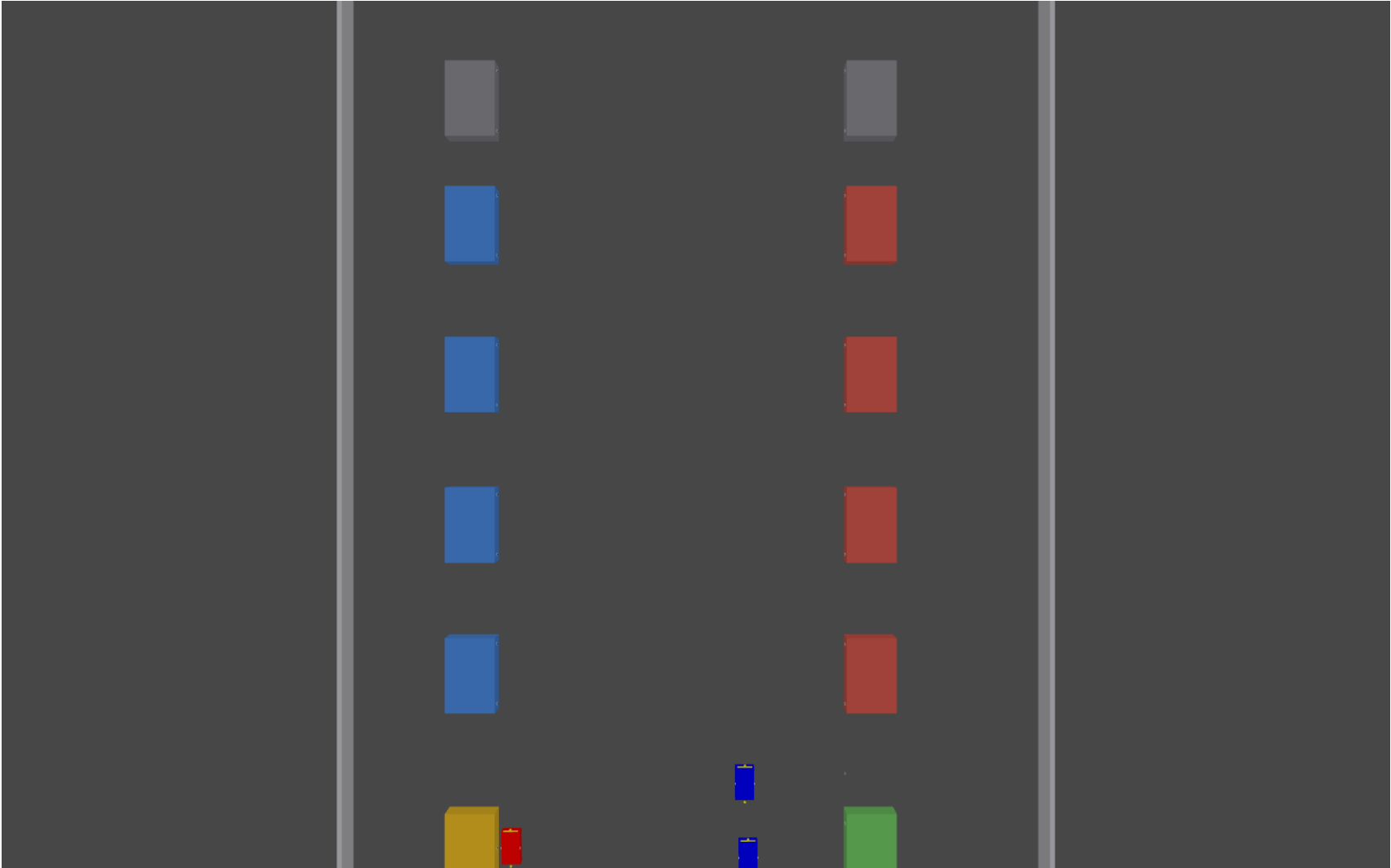


amr1: exit complete

Clear of the bay and the interlock is released. The next robot may come in.

```
amr1 ( -20.40, 7.70) ASRS      exit complete
amr2 ( -15.53, -1.64) CTR3_LD  stood aside, waiting to be passed
amr3 ( -19.20, -7.64) SLT_LD3  machine has not granted entry
amr4 ( -3.36, -7.23) CTR2_ULD  machine has not granted entry
amr5 ( -15.17, -3.61) SLT_LD1  passer: waiting for the yielder to stand aside
17:20:44 STATE amr1(-20.4,+7.7)->ASRS d=1.1 v=0.01 [exit complete] | amr2(-15.5,-1.6)->CTR3_LD d=2.9 v=0.
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rule 1 lower number gives way (1 of 5)

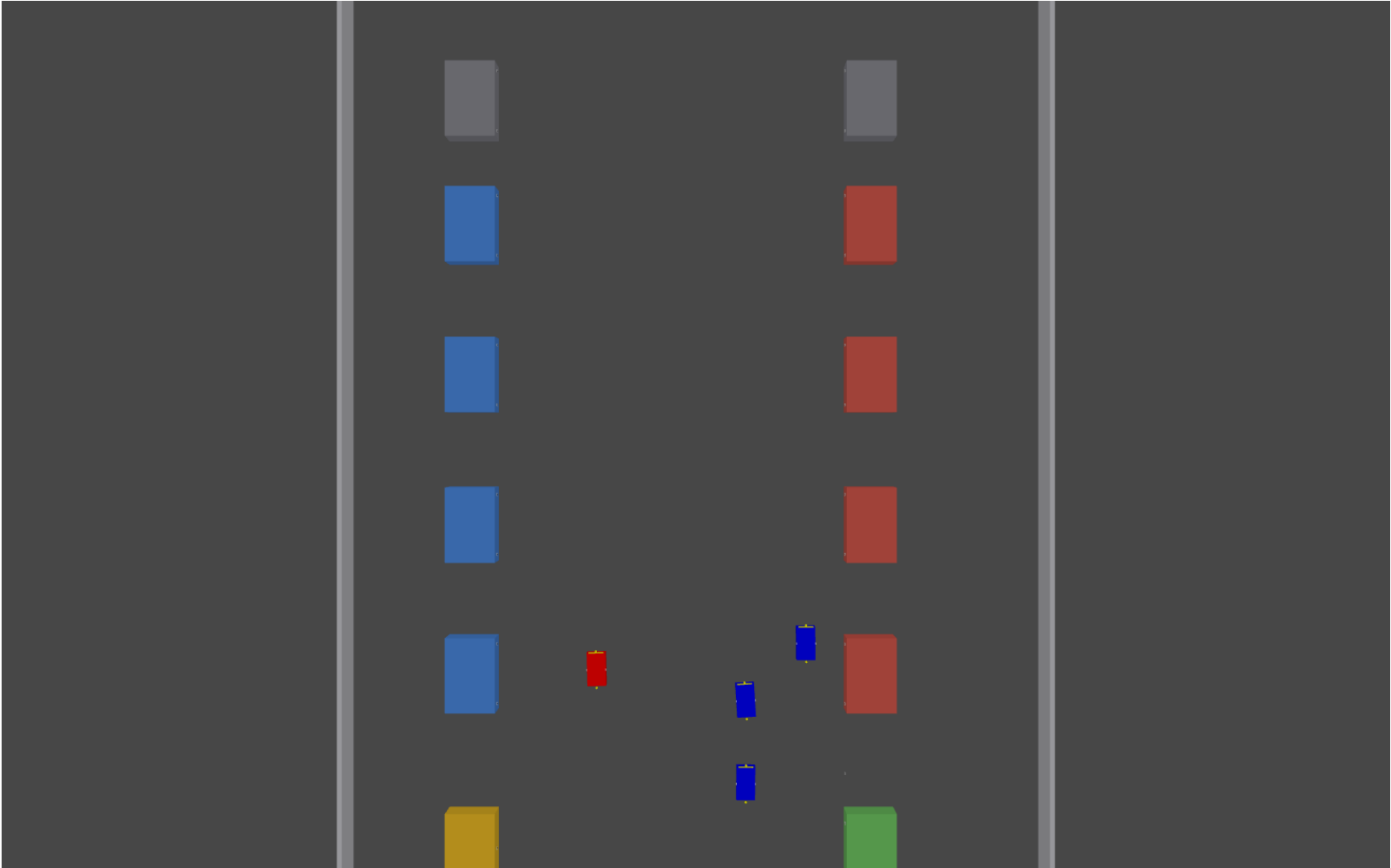


RULE 1 lower number gives way

Two robots met head-on. The lower number always stands aside – a fixed order, so they can never both think they have the road.

amr1	(-20.40, 7.73)	ASRS	dwelling at the port
amr2	(-25.21, 3.40)	GRV1_ULD	dwelling at the port
amr3	(29.01, -1.50)	-	moving
amr4	(-17.18, -3.60)	SLT_LD2	moving
17:18:25 amr4 gives way to amr5			
amr5	(-21.14, -3.74)	CTR1_ULD	passer: waiting for the yielder to stand aside

rule 1 lower number gives way (2 of 5)

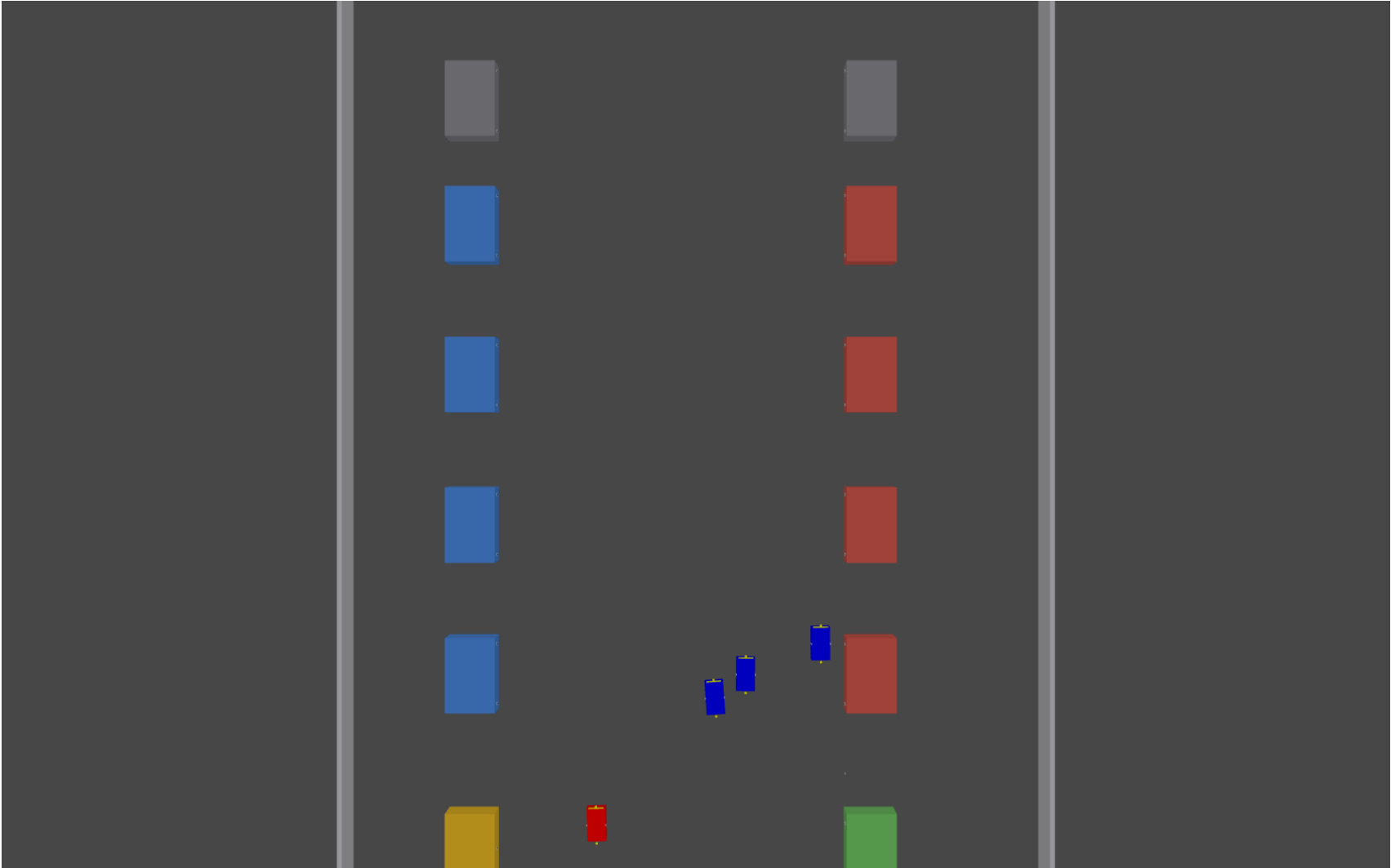


RULE 1 lower number gives way

Two robots met head-on. The lower number always stands aside – a fixed order, so they can never both think they have the road.

amr1	(-11.67, 3.60)	ASRS	exit complete
amr2	(-22.83, 3.60)	CTR3_LD	exit complete
amr3	(-13.09, -3.60)	SLT_LD3	moving
amr4	(-17.52, -3.61)	CTR2_ULD	passer: waiting for the yielder to stand aside
17:19:51 amr5	(-18.55, -6.51)	CTR1_ULD	exit complete

rule 1 lower number gives way (3 of 5)

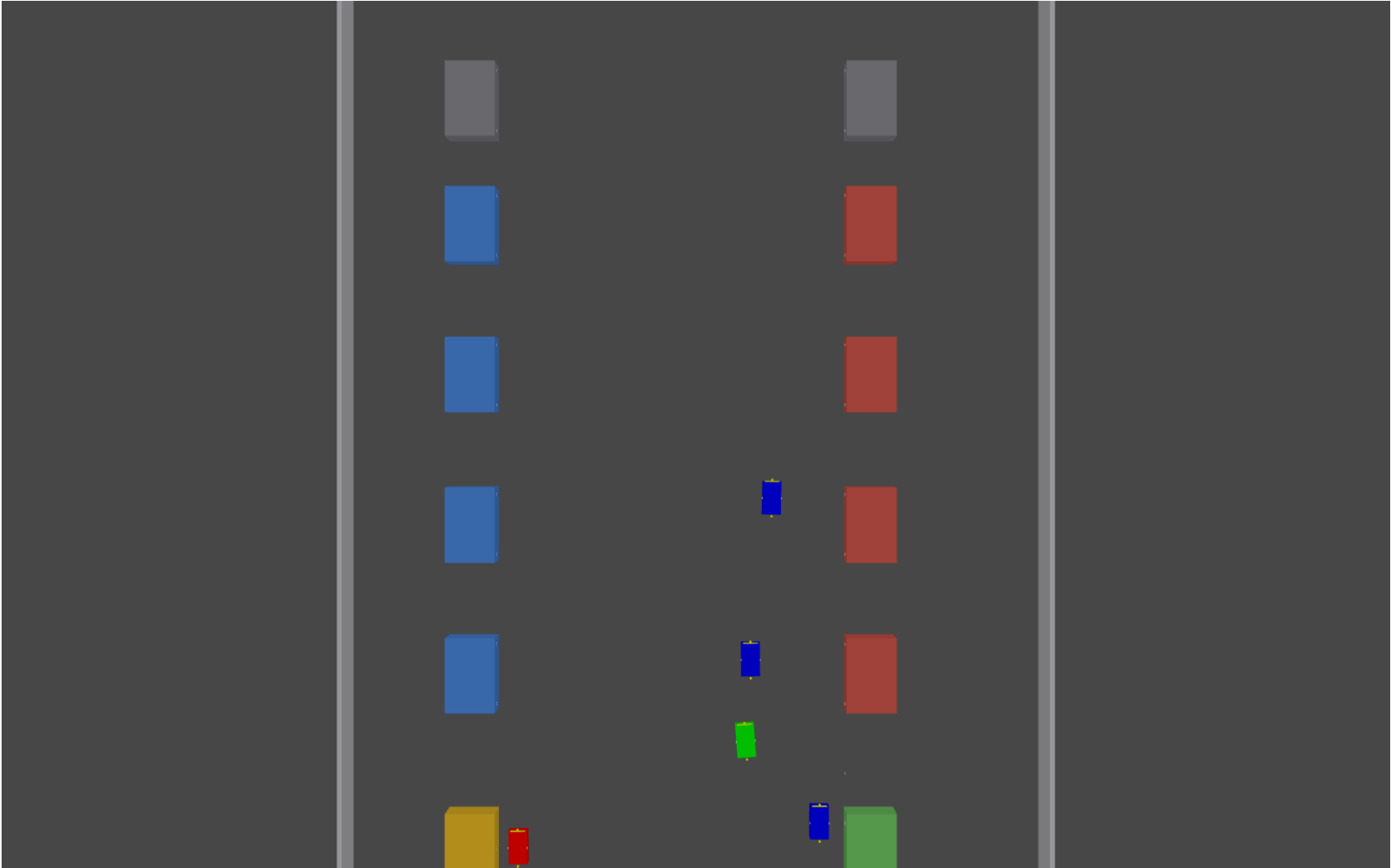


RULE 1 lower number gives way

Two robots met head-on. The lower number always stands aside – a fixed order, so they can never both think they have the road.

amr1	(-19.15,	3.60)	ASRS	exit complete
amr2	(-26.36,	1.42)	CTR3_LD	exit complete
amr3	(-13.16,	-2.04)	SLT_LD3	junction held by another robot
amr4	(-12.10,	-3.60)	CTR2_ULD	passer: waiting for the yielder to stand aside
amr5	(-10.56,	-7.20)	CTR1_ULD	exit complete

rule 1 lower number gives way (4 of 5)



RULE 1 lower number gives way

Two robots met head-on. The lower number always stands aside – a fixed order, so they can never both think they have the road.

amr1	(-20.40, 7.37)	ASRS	exit complete
amr2	(-15.54, -3.60)	CTR3_LD	exit complete
amr3	(-19.20, -7.12)	SLT_LD3	machine has not granted entry
amr4	(-3.54, -4.79)	CTR2_ULD	machine has not granted entry
17:29:34 amr2 gives way to amr3			
amr5	(-11.32, -3.84)	SLT_LD1	passer: waiting for the yielder to stand aside

rule 1 lower number gives way (5 of 5)

RULE 1 lower number gives way

Two robots met head-on. The lower number always stands aside – a fixed order, so they can never both think they have the road.

amr1	(-15.60, 3.69)	GRV3_LD	dwelling at the port
amr2	(-6.02, -3.60)	CTR3_LD	junction held by another robot
amr3	(-17.53, -3.68)	CTR3_ULD	dwelling at the port
amr4	(-3.28, -6.03)		passer: waiting for the yielder to stand aside
amr5	(-23.99, -6.49)	SL1_LD1	passer: waiting for the yielder to stand aside

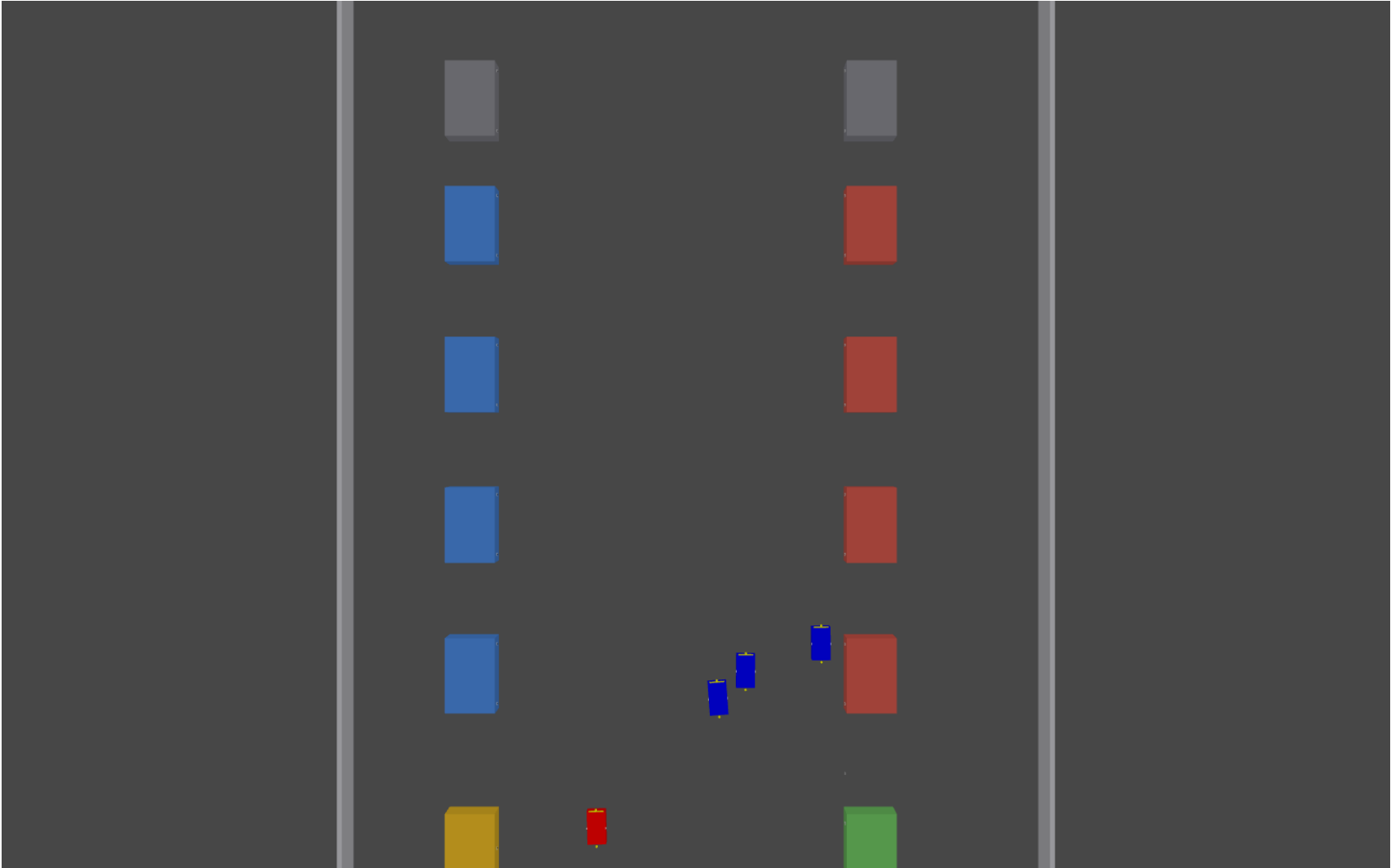
stand aside begun (1 of 5)



stand aside begun (2 of 5)



stand aside begun (3 of 5)



stand-aside begun

The yielder has picked a lay-by clear of every robot it is giving way to, and is driving to it. The other robot holds until it arrives.

```
amr1 ( -19.15,  3.60) ASRS      exit complete
amr2 ( -26.36,  1.42) CTR3_LD  exit complete
amr3 ( -13.16, -2.04) SLT_LD3  junction held by another robot
amr4 ( -12.10, -3.60) CTR2_ULD  passer: waiting for the yielder to stand aside
amr5 ( -10.56, -7.22) CTR1_ULD  exit complete
```

stand aside begun (4 of 5)



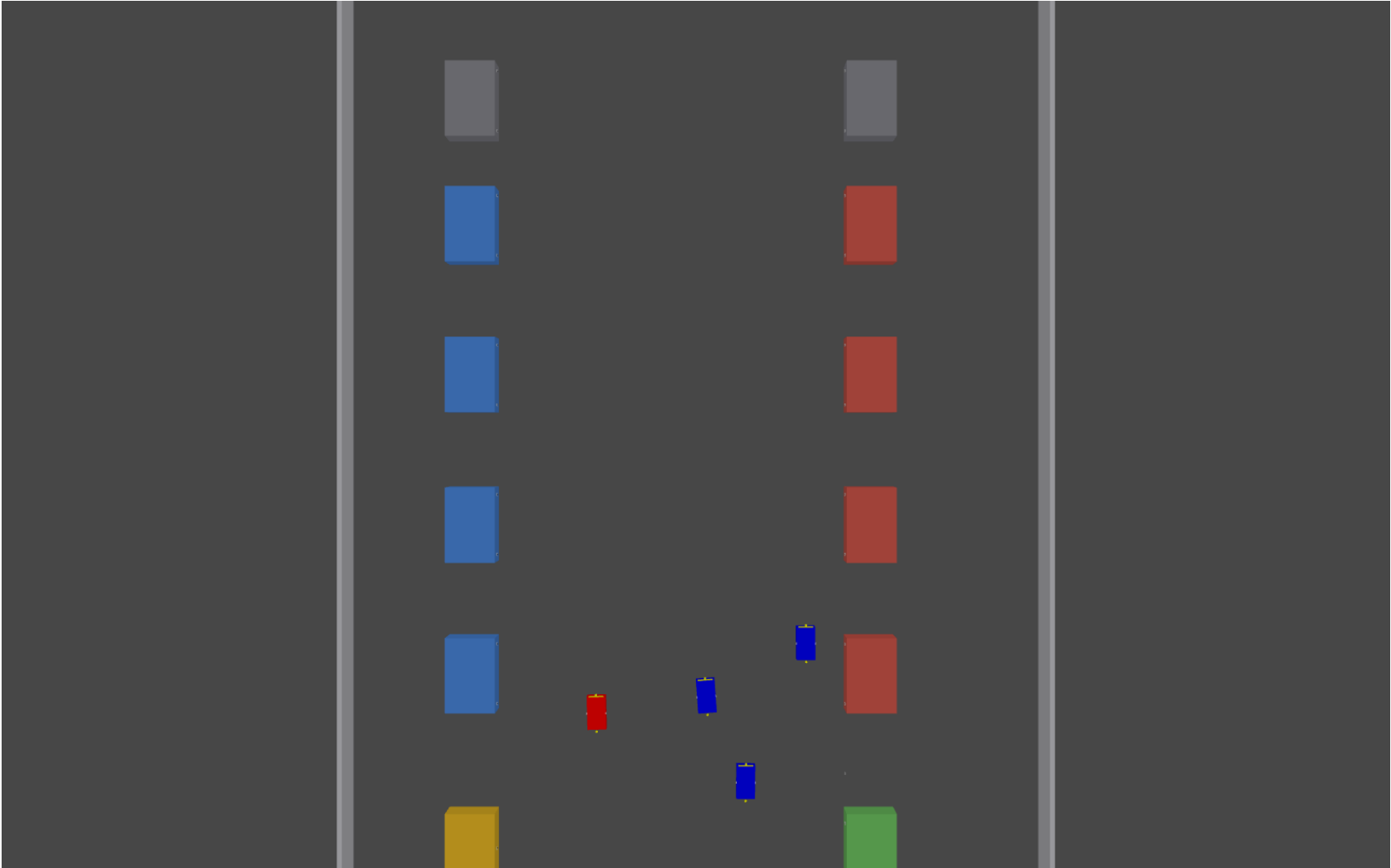
stand aside begun (5 of 5)



stand aside complete (1 of 3)



stand aside complete (2 of 3)



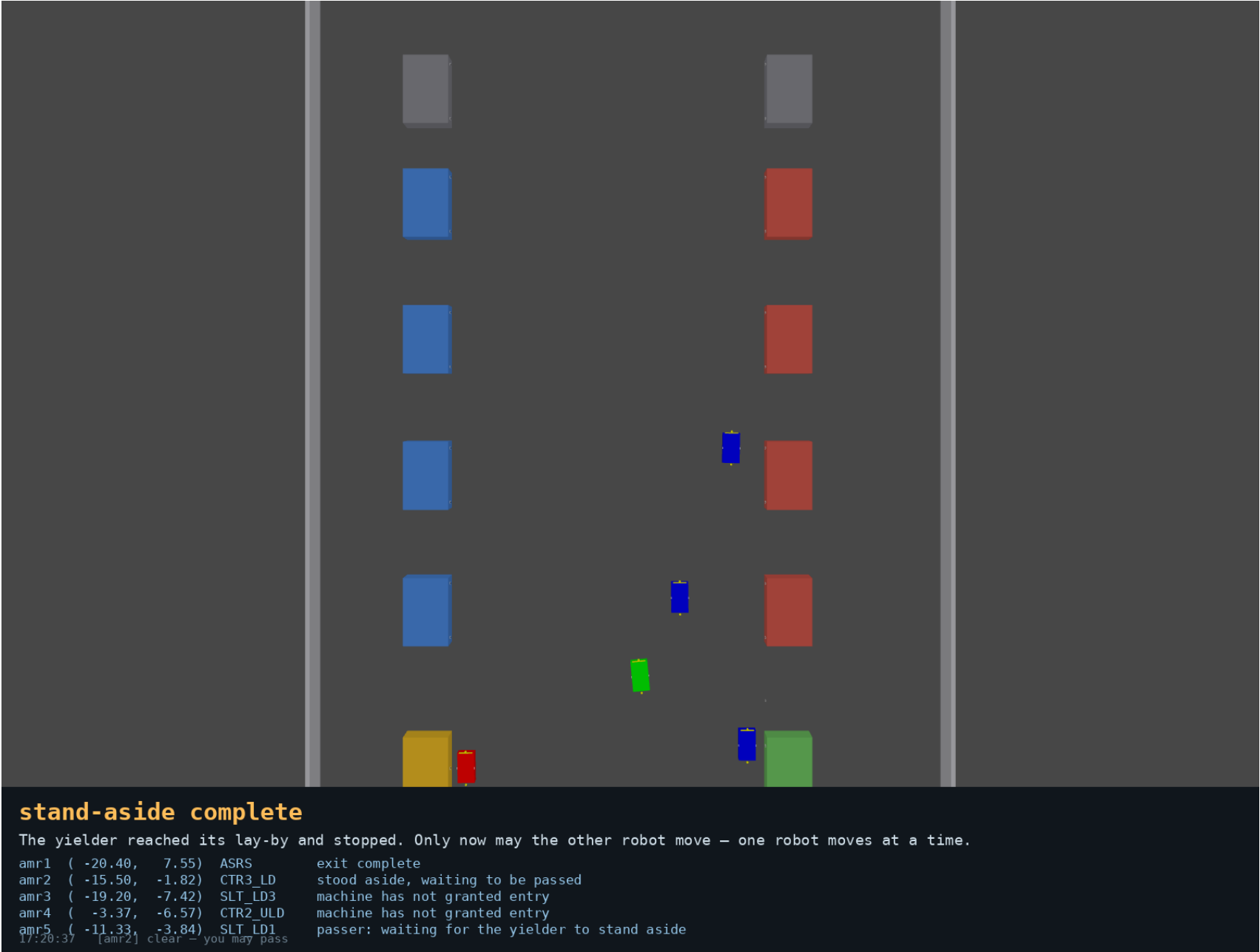
stand aside complete

The yielder reached its lay-by and stopped. Only now may the other robot move – one robot moves at a time.

amr1	(-13.75, 3.60)	ASRS	exit complete
amr2	(-24.91, 3.60)	CTR3_LD	exit complete
amr3	(-13.11, -1.76)	SLT_LD3	stood aside, waiting to be passed
amr4	(-17.25, -3.61)	CTR2_ULD	passer: waiting for the yielder to stand aside
amr5	(-10.55, -6.51)	CTR1_ULD	exit complete

17:19:54 [amr3] clear – you may pass

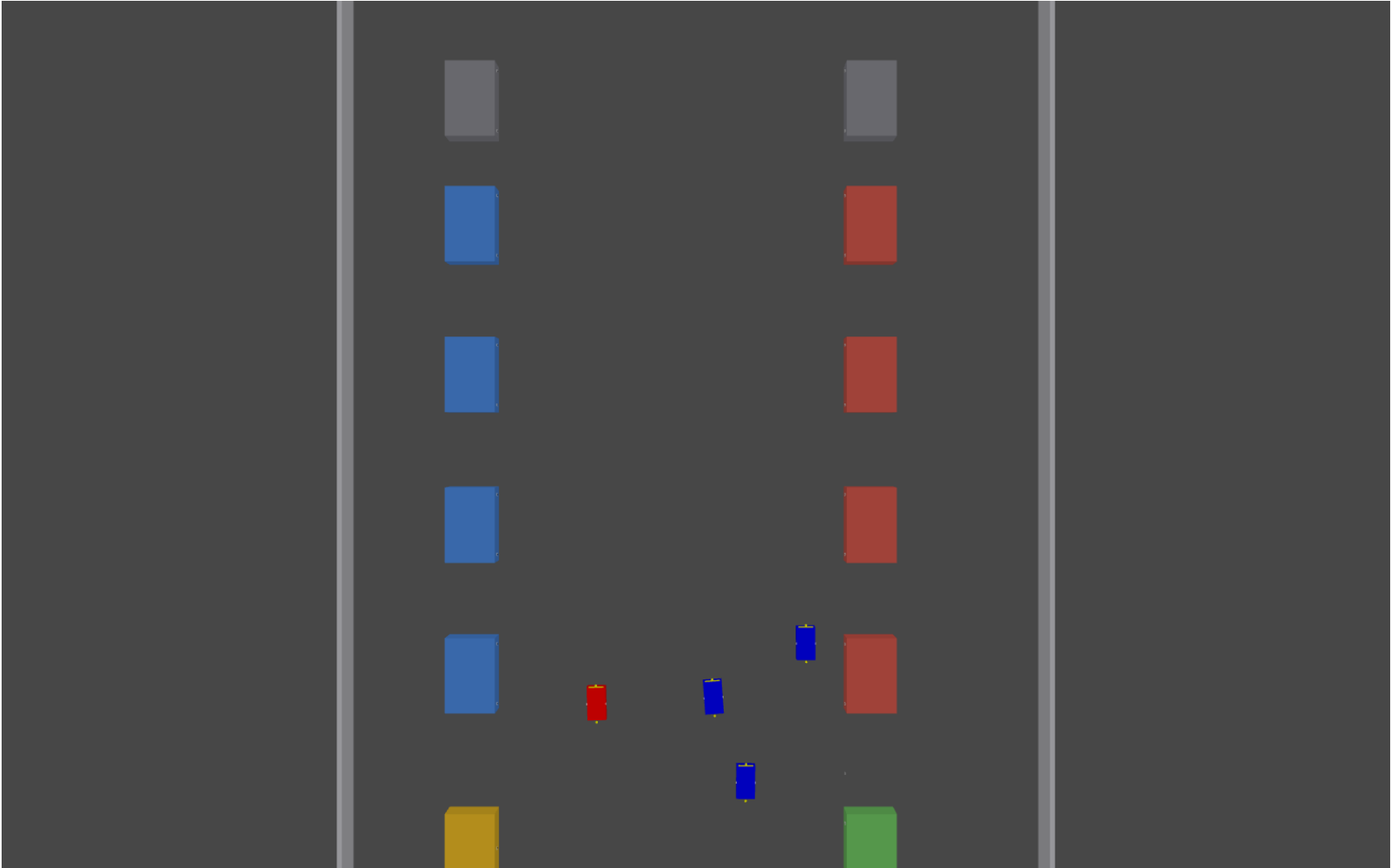
stand aside complete (3 of 3)



passer waiting for the yielder to stand aside (1 of 4)



passer waiting for the yielder to stand aside (2 of 4)



amr4: passer: waiting for the yielder to stand aside

It has right of way but will not move until the other robot says it is clear. One robot moves at a time.

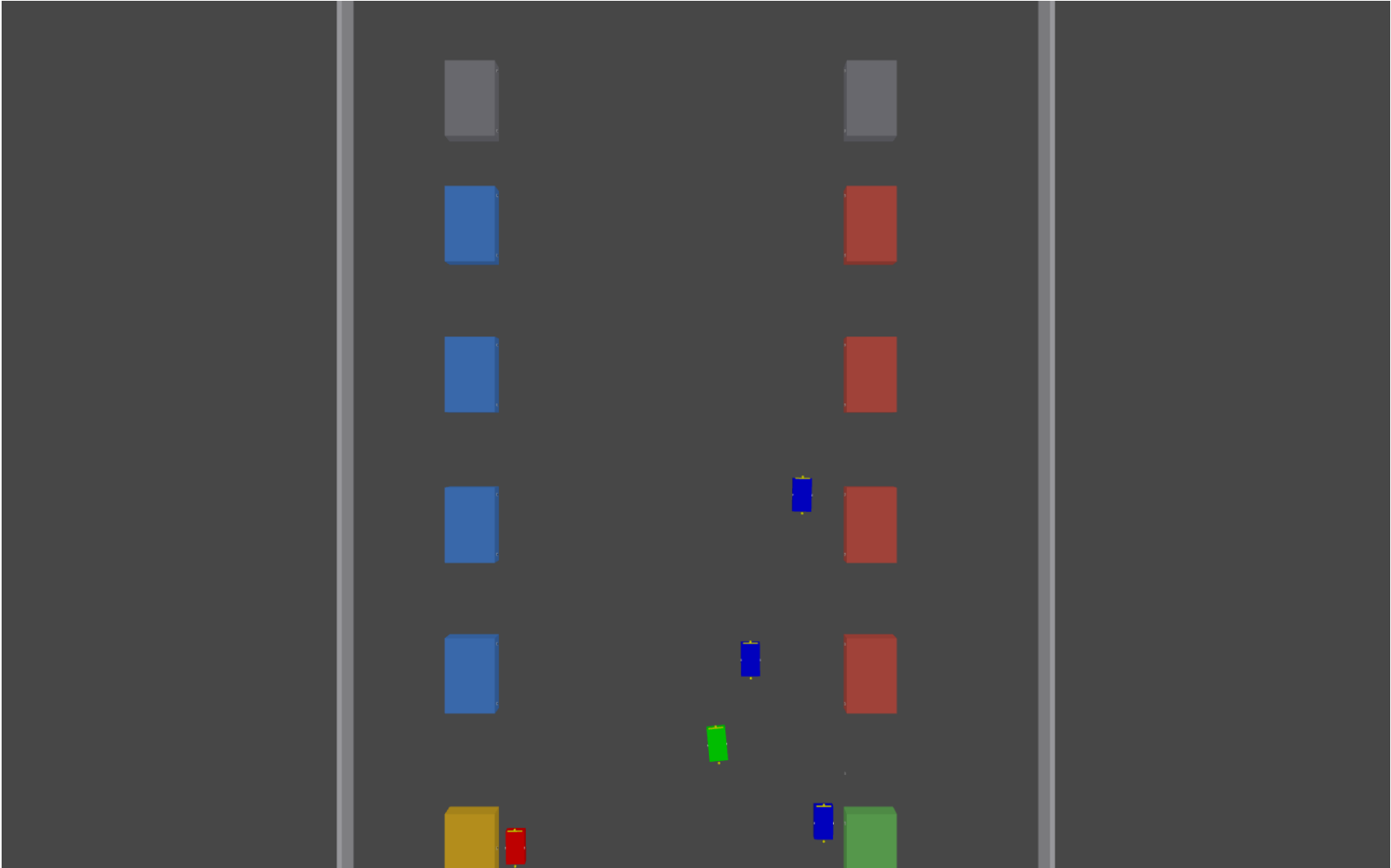
Robot	X	Y	State	Action
amr1	-13.30	3.60	ASRS	exit complete
amr2	-24.48	3.60	CTR3_LD	exit complete
amr3	-13.19	-2.28	SLT_LD3	moving
amr4	-17.25	-3.61	CTR2_ULD	passer: waiting for the yielder to stand aside
amr5	-10.55	-6.51	CTR1_ULD	exit complete

17:19:53 STATE amr1(-13.3,+3.6)=>ASRS d=7.1 v=0.59 | amr2(-24.3,+3.6)->CTR3_LD d=2.1 v=0.59 | amr3(-13.2,

passer waiting for the yielder to stand aside (3 of 4)



passer waiting for the yielder to stand aside (4 of 4)



amr5: passer: waiting for the yielder to stand aside

It has right of way but will not move until the other robot says it is clear. One robot moves at a time.

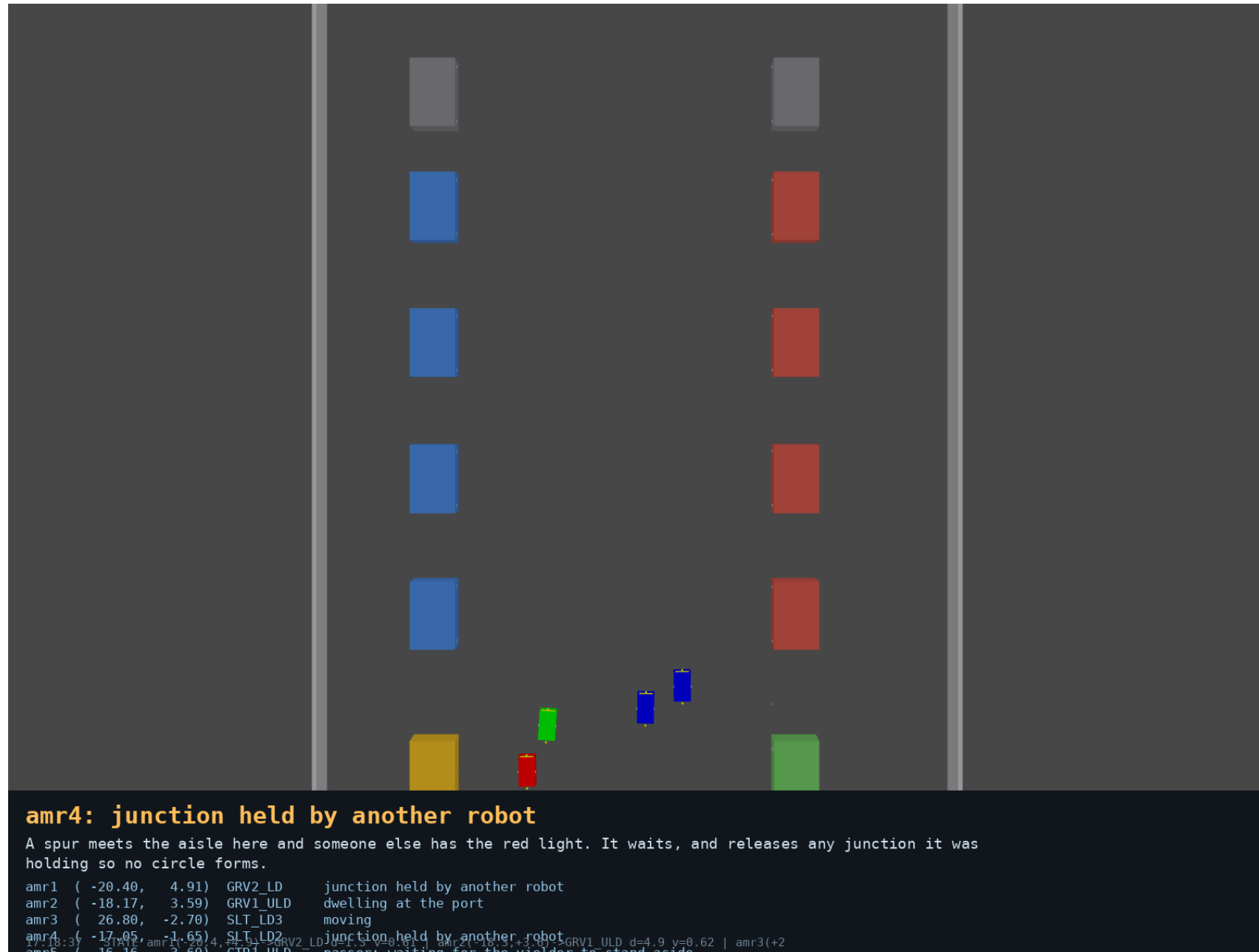
Robot	X	Y	State	Action
amr1	-20.40	7.51	ASRS	exit complete
amr2	-15.40	-2.37	CTR3_LD	exit complete
amr3	-19.20	-7.36	SLT_LD3	machine has not granted entry
amr4	-3.41	-6.11	CTR2_ULD	machine has not granted entry
amr5	-11.33	-3.84	SLT_LD1	passer: waiting for the yielder to stand aside

17:20:36 STATE amr1(-20.4,+7.57)->ASRS d=0.9 v=0.05 | amr2(-15.4,-2.3)->CTR3_LD d=2.2 v=0.54 | amr3(-19.2,

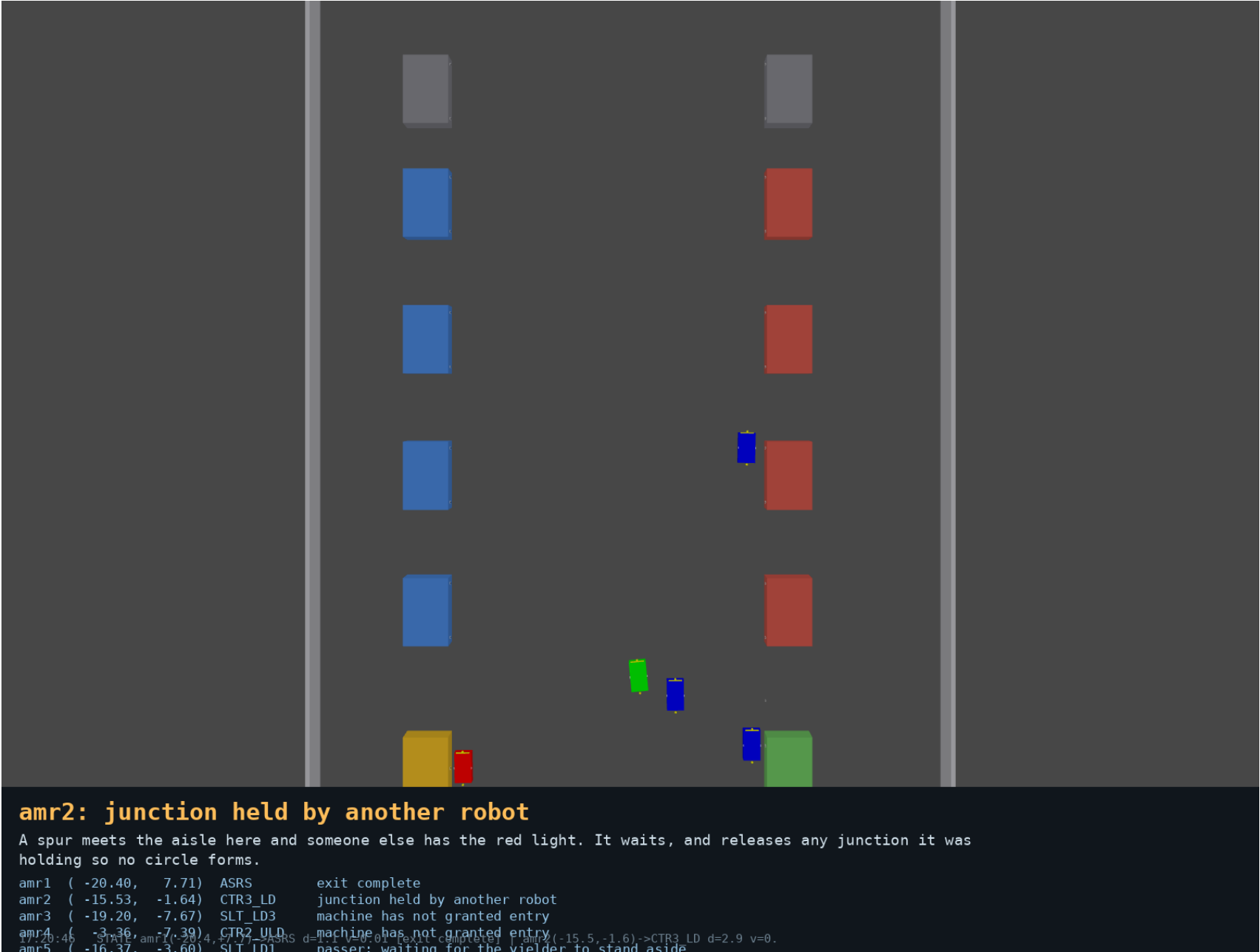
junction held by another robot (1 of 3)



junction held by another robot (2 of 3)



junction held by another robot (3 of 3)



stood aside waiting to be passed (1 of 3)



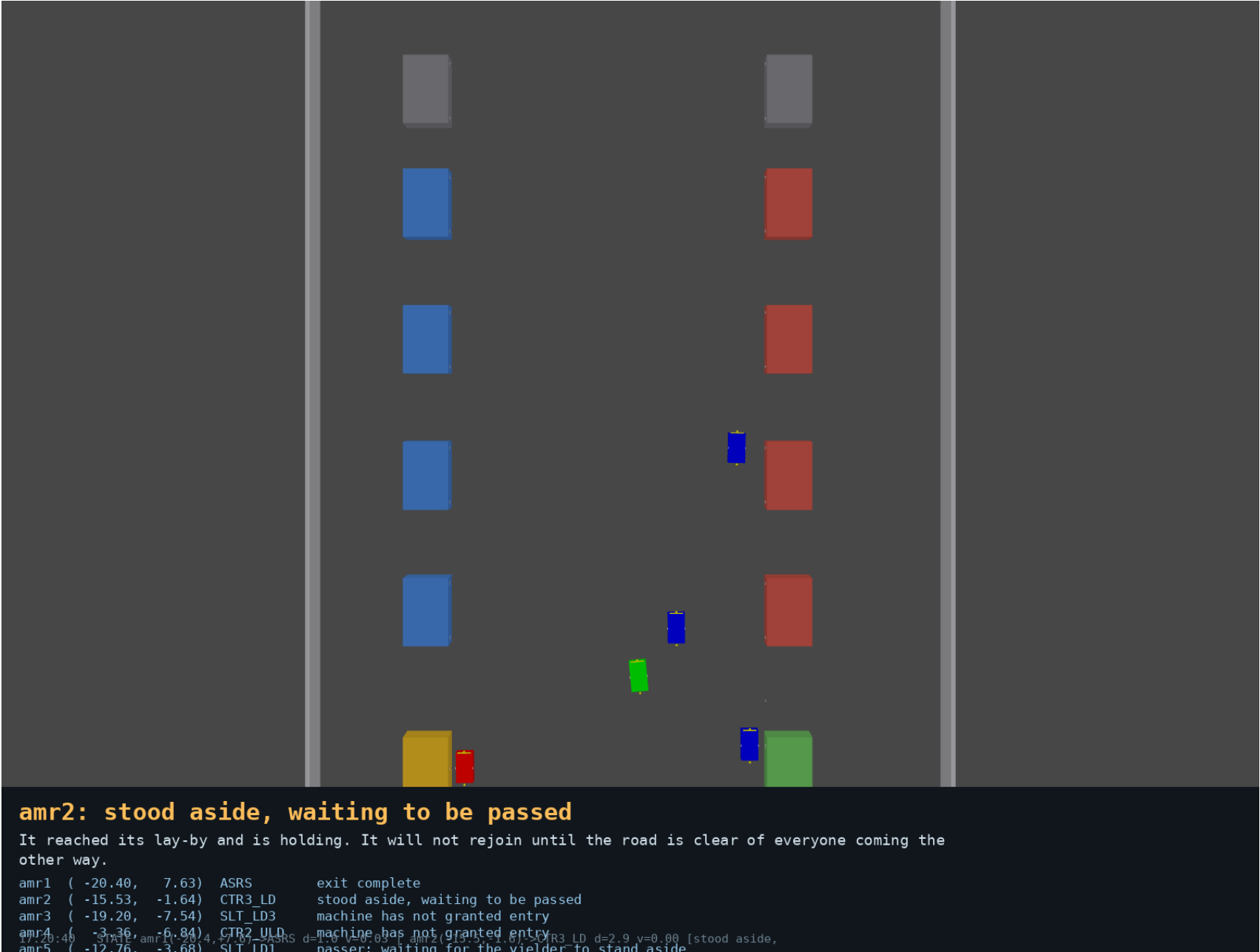
stood aside waiting to be passed (2 of 3)

amr3: stood aside, waiting to be passed

It reached its lay-by and is holding. It will not rejoin until the road is clear of everyone coming the other way.

amr1	(-15.69, 3.60)	ASRS	exit complete
amr2	(-26.07, 3.64)	CTR3_LD	exit complete
amr3	(-13.09, -1.62)	SLT_LD3	stood aside, waiting to be passed
amr4	(-15.71, -3.60)	CTR2_ULD	passer: waiting for the yielder to stand aside
amr5	(-18.55, -6.63)	CTR1_ULD	exit complete

stood aside waiting to be passed (3 of 3)



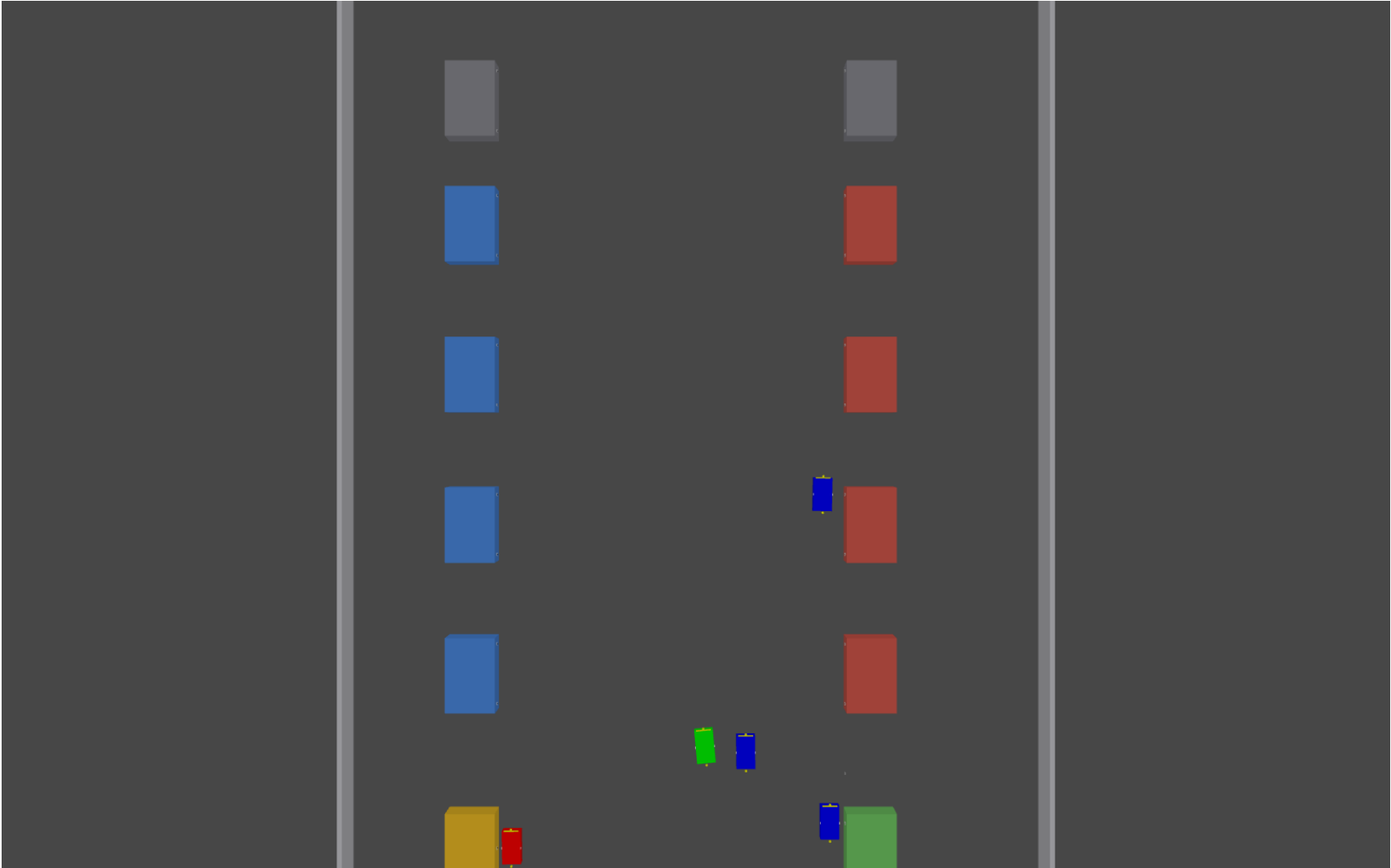
encounter over (1 of 3)



encounter over (2 of 3)



encounter over (3 of 3)



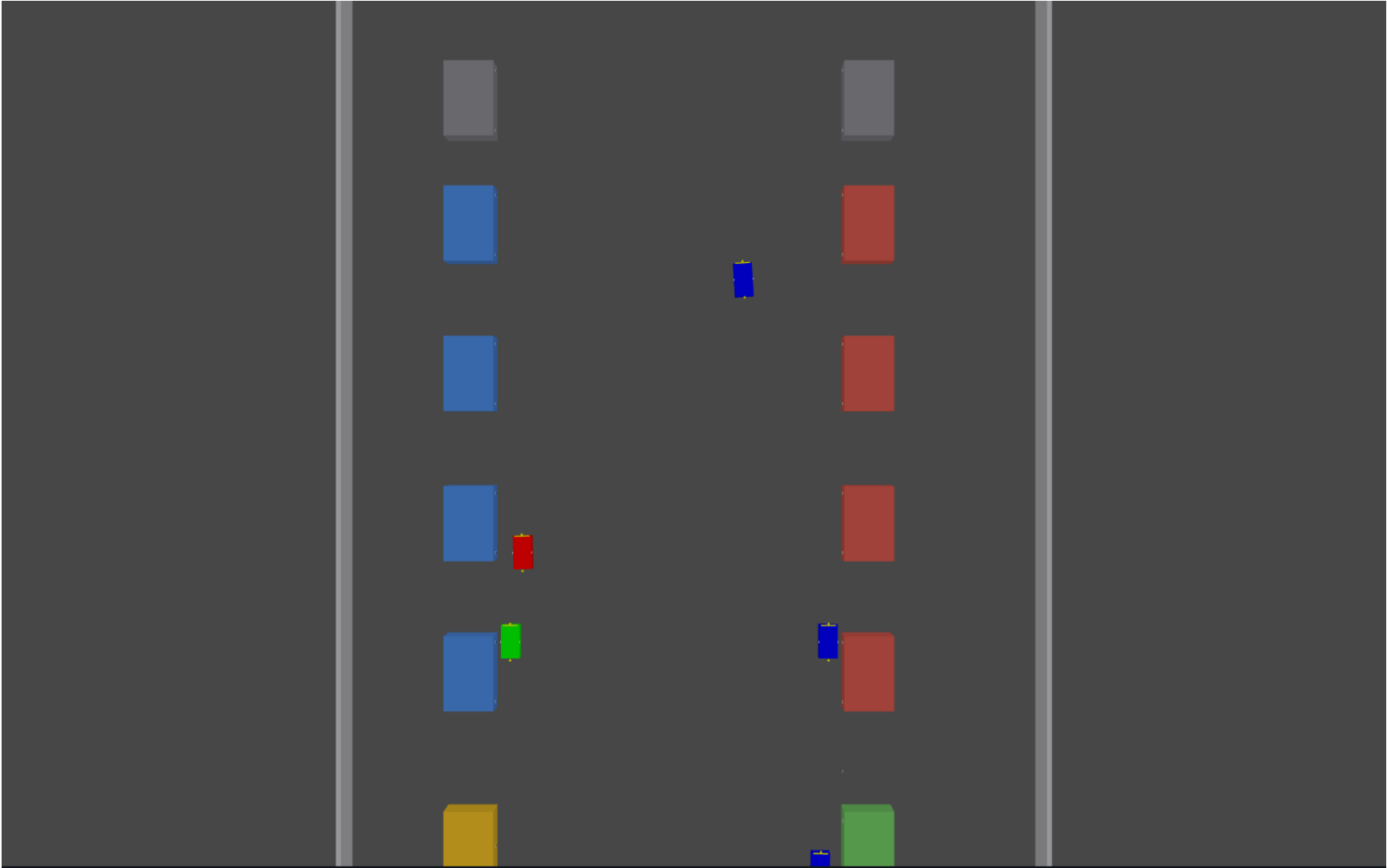
encounter over

Nobody is coming the other way any more. The yielder may rejoin the lane and carry on.

amr1	(-20.40,	7.70)	ASRS	exit complete
amr2	(-15.53,	-1.64)	CTR3_LD	stood aside, waiting to be passed
amr3	(-19.20,	-7.66)	SLT_LD3	machine has not granted entry
amr4	(-3.36,	-7.30)	CTR2_ULD	machine has not granted entry
amr5	(-15.62,	-3.61)	SLT_LD1	passer: waiting for the yielder to stand aside

17:20:45 [amr5] road is clear = it is yours

layer 1 on course to touch another robot (1 of 1)



amr2: layer 1: on course to touch another robot

The last safety check before any wheel command. Sampling both bodies 2 s ahead says they would come within 0.30 m. It only ever says stop – deciding who goes is a rule above it.

```
amr1 ( -6.24, 7.06) GRV2_LD machine has not granted entry
amr2 ( -10.56, 7.68) GRV1_ULD layer 1: on course to touch another robot
amr3 ( 7.16, -3.60) SLT_LD3 moving
amr4 ( -21.60, -7.31) SLT_LD2 machine has not granted entry
17.19:16 STATE amr1 (-6.24, 7.06) GRV2_LD v=0.5 v=0.10 | amr2 (-10.56, 7.68) GRV1_ULD d=1.1 v=0.02 [layer 1:
amr5 ( -10.56, -7.70) CTR1_ULD machine has not granted entry
```